

Systemic Tau and Hierarchical Ordinal Conjunctions: A Relational Theory of Critical Transitions*

Johel Padilla-Villanueva

Department of Environmental Health, University of Puerto Rico

johelpadilla@gmail.com

August 1, 2026

Abstract

Complex systems frequently undergo critical transitions—abrupt reorganizations of macroscopic dynamics under small changes in drivers or connectivity. Classical early warning signals (EWS) based on variance, lag-1 autocorrelation, and related signatures of critical slowing down are powerful yet primarily univariate, magnitude-driven, and tied to a narrow class of local bifurcations. In high-dimensional, nonstationary settings they are often weak, delayed, or directionally ambiguous: fluctuations may grow while relational organization tightens, or vice versa.

We present foundations for a dual, relational account based on two complementary constructs. **Systemic Tau** (τ_s) is an ordinal, multivariate measure of cross-variable coupling: in sliding windows it aggregates Kendall-type rank concordance into $\tau_s(t) \in [-1, 1]$, tracking relational coherence rather than the amplitude of any single observable. The **Discrete Extramental Clock** (RECD; *Reloj Extramental Discreto*)—the operational scaffolding of the nested hierarchy—supplies an intrinsic, event-driven temporal metric built from hierarchical *ordinal conjunctions*. From Bandt–Pompe symbolic dynamics we define three levels of structural claim—coincidence (Φ_1), persistent relational structure (Φ_2), and the irreducible residual beyond pairwise models (Res_{pair})—and a scalable hybrid proxy (excess3) that is *not* identical to Res_{pair} . The discrete clock advances as $\Delta\text{RECD}(t) = \sum_{\ell=1}^3 \alpha_\ell(\lambda) \widehat{M}_\ell(t)$, where $\widehat{M}_\ell$ are level estimators and $\alpha_\ell(\lambda)$ are admissible *aggregation weights* modulated by a regime cue λ (typically derived from $|\tau_s|$ and calibrated with Feigenbaum-motivated scales). Systemic Tau asks whether mutual organization is reorganizing; nested RECD asks *at what ordinal depth* that reorganization is expressed. We develop the mathematics of both objects, the status of the aggregation policy (λ, α) , the operational link to the Feigenbaum route, and a brief synthetic validation on coupled logistic maps. Early warning is thereby cast as a change in hierarchical ordinal structure, not merely a rise in fluctuation size.

Keywords: Systemic Tau, RECD, ordinal patterns, critical transitions, regime shifts, early warning signals, complex systems, Feigenbaum universality, hierarchical ordinal conjunctions

*Preprint DOI (this version): <https://doi.org/10.5281/zenodo.21753560>. Concept DOI (all versions): <https://doi.org/10.5281/zenodo.21287251>. Source and formal notes: <https://github.com/johelpadilla/systemic-tau-recd-foundations>.

1 Introduction

Complex systems across the physical, biological, ecological, and social sciences exhibit a striking commonality: they can remain for long periods in a macroscopic regime and then reorganize abruptly when a control parameter, external driver, or internal connectivity pattern crosses a threshold. Such events—tipping points, regime shifts, or critical transitions—are often difficult to reverse once completed, and their anticipation remains a central problem of nonlinear science Scheffer et al. [2009], Scheffer [2012], Dakos et al. [2012]. The scientific challenge is not merely to register the transition after it has occurred, but to detect structural precursors while the system is still formally within the pre-critical attractor.

Over the past two decades, the dominant theoretical response has been the theory of *critical slowing down* (CSD). Near certain local bifurcations, the dominant eigenvalue of the Jacobian approaches the unit circle, recovery rates from perturbations decline, and generic statistical signatures appear: rising variance, increasing lag-1 autocorrelation, changes in skewness, and related indicators collectively known as early warning signals (EWS) Scheffer et al. [2009], Dakos et al. [2012], Boettiger et al. [2013]. These tools have organized an extensive empirical programme in ecology, climate science, finance, and physiology. Their conceptual elegance, however, rests on assumptions that are frequently violated outside low-dimensional, quasi-stationary laboratory or model systems.

First, classical EWS are predominantly *univariate* and *magnitude-driven*: they monitor the fluctuation statistics of a single observable (or of several observables treated independently). In genuinely high-dimensional systems the relevant information often resides not in the amplitude of any one coordinate, but in the *pattern of mutual organization* among many coordinates—how ranks, orderings, and symbolic relations cohere or dissolve. Second, CSD-type indicators are tuned to a comparatively narrow class of local bifurcations; many empirically important transitions are topological, global, or mediated by intermittent, nonstationary, or noise-dominated dynamics for which variance and autocorrelation are weak, delayed, or directionally reversed Boettiger et al. [2013], Dakos et al. [2015], Ditlevsen and Johnsen [2010]. Third, chronological sampling itself is an external clock: uniform observation intervals do not necessarily coincide with the intrinsic “ticks” at which the system updates its relational structure. A fuller critique of these limitations is developed in Section 2.

That language must be *relational* and *ordinal*. Relational, because critical transitions frequently reorganize couplings among variables before any single series reaches an extreme value. Ordinal, because rank-order and permutation-based encodings are invariant under monotone transformations, comparatively robust to moderate outliers, and naturally aligned with symbolic dynamics Bandt and Pompe [2002], Amigó [2010], Keller et al. [2014]. Ordinal methods—permutation entropy, ordinal networks, recurrence quantification on symbolic series—have already demonstrated that much of the dynamical signature of nonlinearity lives in the geometry of relative ordering rather than in absolute scale. What has been missing is a coherent theory that (i) quantifies the evolution of multivariate rank coherence as a stability metric and (ii) constructs from hierarchical ordinal events an intrinsic discrete time whose mathematical regime itself changes near criticality.

The present paper supplies that theory through two interlocking constructs.

Systemic Tau. **Systemic Tau** (τ_s) is an ordinal, multivariate measure of cross-variable coupling dynamics. Given a d -variate series $X(t) = (X_t^{(1)}, \dots, X_t^{(d)})$, one computes in each sliding window of

length W an aggregated Kendall-type concordance among coordinates, yielding a scalar $\tau_s(t) \in [-1, 1]$. Values near $+1$ indicate tight mutual rank organization; values near -1 indicate strong anti-concordance; intermediate values mark the loss or inversion of relational structure. Operational thresholds—in particular a stability band $\tau_s \geq 0.50$, a chaotic band $|\tau_s| < 0.41$, and an anti-synchronization band $\tau_s \leq -0.41$ —are Feigenbaum-motivated calibrations rather than *ad hoc* percentiles of a single dataset Feigenbaum [1978, 1983]; they are design choices, not theorems of the period-doubling cascade (Section 6). The directed contrast

$$\Delta\tau_s = \overline{\tau}_s^{(\text{approach})} - \overline{\tau}_s^{(\text{basal})}$$

then measures reorganization of coupling, not growth of univariate fluctuation. Systemic Tau answers the question: *is the pattern of mutual organization among variables changing as a transition approaches?*

The RECD framework. The **Discrete Extramental Clock** (RECD; from the Spanish *Reloj Extramental Discreto*) addresses a complementary question: *when*, and *at what depth*, do multivariate ordinal events cohere strongly enough to advance a system-generated time index? Chronological time is treated as an external imposition; RECD replaces it with an intrinsic, event-driven metric built from hierarchical *ordinal conjunctions*. Each coordinate is encoded by Bandt–Pompe ordinal patterns $\pi_t^{(i)}$ of embedding dimension m and delay τ . Conjunctions are not a single flat rule but a nested hierarchy:

- **Level 1 — coincidence** (Φ_1): simultaneous equality (or co-occurrence) of ordinal symbols across coordinates—the weakest, purely statistical layer of ordinal synchronization.
- **Level 2 — persistent relational structure** (Φ_2): relations among patterns (lead/lag, opposition, fixed joint transitions, meta-ordinal configurations) that persist for at least $p_{\min}$ consecutive steps, adding memory and relational quality beyond mere coincidence.
- **Level 3 — irreducible residual** (strong claim Res_{pair} ; operational proxy excess3): joint ordinal configurations not reproducible by pairwise (order- ≤ 2) interaction models. The structural target is the information-projection residual Res_{pair} ; the continuous hybrid excess3 used in most numerical sections is a *proxy*, not an identification of that residual (Section 4.3).

These contributions combine into a depth-dependent update of the discrete clock,

$$\Delta\text{RECD}(t) = \alpha_1(\lambda) \widehat{M}_1(t) + \alpha_2(\lambda) \widehat{M}_2(t) + \alpha_3(\lambda) \widehat{M}_3(t), \quad (1)$$

where $\widehat{M}_1 = \Phi_1$, $\widehat{M}_2 = \Phi_2$, and $\widehat{M}_3$ is either excess3, binary Φ_3 , or Res_{pair} when estimated. Nesting is of *structural claims*, not of the binary indicators themselves (Section 4.2): deeper claims demand strictly richer relational structure and are not reducible to shallower ones. The coefficients α_ℓ need not be constant: a regime cue λ , typically a calibrated transform of $|\tau_s|$ with Feigenbaum-motivated scales, reweights the hierarchy so that near chaos deeper layers acquire greater *aggregation share* of ΔRECD . That reweighting is an aggregation policy, not a new geometric observable of the joint symbol law. The RECD therefore does not advance homogeneously: its speed, irregularity, and jump structure reflect both active conjunction depth and the chosen weights. Systemic Tau and RECD are dual: the former detects directed relational reorganization; the latter localizes that reorganization on a hierarchical ordinal ladder and converts it into intrinsic discrete time.

Contributions and organization. The main contribution of this article is foundational rather than applied. We provide a rigorous, self-contained foundation for Systemic Tau and hierarchical ordinal conjunctions—operationalized through the nested RECD—as a unified relational theory of critical transitions in complex systems. Specifically, we:

1. articulate the conceptual and mathematical limitations of classical EWS that motivate a shift from magnitude-based to ordinal–relational observables (Section 2);
2. formalize Systemic Tau as a multivariate measure of coupling dynamics, including its operational thresholds and interpretation near bifurcations (Section 3);
3. develop the RECD hierarchy of ordinal conjunctions—Levels 1–3, nesting, and the irreducible residual Res_{pair} versus its proxy excess3 (Section 4);
4. introduce weighted RECD through the aggregation schedule $\alpha(\lambda)$, with a sharp separation of Level 3 abundance from the aggregation share f_ℓ (Section 5);
5. situate operational thresholds and cue scales within the Feigenbaum route, distinguishing inherited universality from operational analogy (Section 6);
6. report a brief synthetic validation on coupled logistic maps that contrasts proxy abundance, estimation of the irreducible residual, and regime-aware aggregation share (Section 7); and
7. discuss conceptual implications, limitations, and outlook (Sections 8–9).

Empirical applications to epidemiology, cardiac dynamics, and related domains are cited only as motivating context. The paper’s thesis is that critical transitions can be formalized as reorganizations of hierarchical ordinal structure, and that Systemic Tau together with RECD provide a coherent language in which that thesis can be stated, tested, and refined—including an explicit account of what is claim, what is estimator, and what is aggregation policy.

2 Limitations of Classical Early Warning Signals

The classical early-warning programme rests on a precise dynamical scenario. Consider a dynamical system of low dimension

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}; \mu), \quad \mathbf{x} \in \mathbb{R}^n, \quad (2)$$

with a stable fixed point $\mathbf{x}_*(\mu)$ that loses stability as a control parameter μ approaches a critical value μ_c . Near a local bifurcation at which the real part of the dominant eigenvalue of the Jacobian $D\mathbf{f}$ tends to zero, the linear recovery rate from small perturbations vanishes. In the presence of additive noise this *critical slowing down* (CSD) produces generic statistical signatures in a monitored scalar observable $y(t)$: rising variance, increasing lag-1 autocorrelation, and, under further assumptions, changes in skewness or spectral reddening Scheffer et al. [2009], Dakos et al. [2012], Kuehn [2011]. These indicators have organized an extensive empirical literature and remain indispensable when the underlying assumptions are approximately satisfied.

The difficulty is that many systems of scientific interest violate those assumptions simultaneously. We group the resulting limitations into five interlocking classes.

Univariate and magnitude-driven character. Standard EWS track the amplitude statistics of one series, or of several series treated as independent monitors. Critical transitions in coupled systems, however, frequently reorganize *cross-variable* structure before any single coordinate exhibits extreme fluctuations Dakos et al. [2015], Boettiger et al. [2013]. Two coordinates may remain bounded while their rank concordance collapses; conversely, a single series may become more variable while multivariate organization tightens. Univariate variance and autocorrelation are blind to this distinction: they answer whether a series fluctuates more, not whether the relational geometry of the state space is changing.

Narrow bifurcation class. CSD theory is most rigorous for local bifurcations of fixed points (or analogous slowings of low-dimensional attractors). Many empirically important transitions are global, topological, or mediated by basin boundary collisions, intermittency, or network rewiring for which the dominant eigenvalue picture is incomplete or inapplicable Kuehn [2011], Ditlevsen and Johnsen [2010]. In such cases the “generic” CSD signature may be weak, absent, or appear only after the transition has effectively begun.

Nonstationarity, noise, and short records. Ecological, physiological, and socio-technical time series are typically short, nonstationary, and contaminated by measurement and process noise. Estimators of lag-1 autocorrelation and variance are sensitive to trends, heteroscedasticity, and window choice; detrending procedures intended to stabilize the estimates can themselves inject or erase apparent early warnings Dakos et al. [2012], Boettiger et al. [2013]. Magnitude-based indicators also respond to pure scale changes (e.g., sensor gain, seasonal amplitude) that need not correspond to structural reorganization.

Directional ambiguity and false positives. Even when variance or autocorrelation rise, the sign of the change need not encode a unique physical interpretation. Rising variance can accompany both loss of resilience and beneficial increases in exploratory fluctuation; falling autocorrelation can coincide with either stabilization or the onset of high-dimensional mixing. Without a relational reference, classical EWS are underdetermined as statements about *how* the system is reorganizing Boettiger et al. [2013], Dakos et al. [2015].

External time versus intrinsic event structure. Finally, classical EWS are indexed by chronological sampling time. The observation grid is an external clock. Near criticality, however, the relevant events may be sparse relational updates—moments at which several degrees of freedom cohere or decorrelate—rather than uniform increments of laboratory time. An indicator that only inspects amplitude on a fixed grid can miss the discrete structure of those updates.

None of these limitations invalidates CSD theory within its domain of validity. They do imply that a complementary language is needed for high-dimensional, noisy, and nonstationary systems: one that (i) measures multivariate relational change rather than univariate amplitude, (ii) remains robust under monotone rescaling, and (iii) admits an intrinsic, event-driven temporal metric. Systemic Tau and the RECD framework are constructed precisely for that purpose.

3 Systemic Tau: A Relational Measure of Coupling Dynamics

3.1 Definition

Let

$$X(t) = (X_t^{(1)}, \dots, X_t^{(d)}) \in \mathbb{R}^d, \quad t = 1, \dots, T, \quad (3)$$

be a d -variate time series with $d \geq 2$. Fix a sliding window length $W \geq 3$ and, at each time $t \geq W$, consider the windowed matrix

$$\mathcal{W}_t = \{X(t - W + 1), \dots, X(t)\} \in \mathbb{R}^{W \times d}. \quad (4)$$

For each coordinate $i = 1, \dots, d$, let $r_t^{(i)} \in \mathbb{R}^W$ denote the rank transform of the i -th column of $\mathcal{W}_t$ (mid-ranks for ties). The Kendall rank correlation between coordinates i and j in the window is

$$\tau_K(r_t^{(i)}, r_t^{(j)}) = \frac{C_{ij} - D_{ij}}{\binom{W}{2}}, \quad (5)$$

where C_{ij} and D_{ij} are the numbers of concordant and discordant pairs among the W observations Kendall [1938]. **Systemic Tau** is the mean pairwise concordance

$$\tau_s(t) = \frac{2}{d(d-1)} \sum_{1 \leq i < j \leq d} \tau_K(r_t^{(i)}, r_t^{(j)}) \in [-1, 1]. \quad (6)$$

Equivalently, $\tau_s(t)$ is the average Kendall association across all unordered pairs of coordinates. The construction is ordinal by design: any componentwise strictly increasing transformation of X leaves τ_s invariant, and moderate outliers affect ranks only locally.

Several remarks clarify the interpretation.

1. Values $\tau_s(t) \rightarrow +1$ indicate tight mutual rank organization (high ordinal coherence across variables).
2. Values $\tau_s(t) \rightarrow -1$ indicate systematic anti-concordance (anti-synchronization of ranks).
3. Intermediate values mark partial loss, fragmentation, or inversion of relational structure.
4. τ_s is *not* a univariate complexity score: it is undefined (or trivial) for $d = 1$, and its content is irreducibly cross-variable.

3.2 Operational thresholds and Feigenbaum-motivated bands

Empirical and theoretical calibration within the Systemic Tau programme identifies three operational bands for τ_s , motivated by Feigenbaum universality rather than by free fitting to a particular dataset Feigenbaum [1978, 1983], Padilla-Villanueva [2026b]:

$$\begin{aligned} \tau_s &\geq \tau_{\text{st}} := 0.50 && \text{(stability / strong ordinal synchronization),} \\ |\tau_s| &< \tau_{\text{ch}} := 0.41 && \text{(genuine chaotic / high-volatility band),} \\ \tau_s &\leq -\tau_{\text{ch}} = -0.41 && \text{(strong anti-synchronization).} \end{aligned} \quad (7)$$

The chaotic threshold $\tau_{\text{ch}} \approx 0.41$ is numerically consistent with the reciprocal structure associated with Feigenbaum’s constant $\delta \approx 4.6692$ (Section 6); the stability threshold $\tau_{\text{st}} = 0.50$ marks the regime in which rank coherence is strong enough that structural “ticks” of an extrinsic clock contribute little new relational information. These bands are *operational*: they organize interpretation and weighting (Section 5) and should be re-validated when the observational proxy or embedding changes substantially. They are not claimed as sharp mathematical bifurcations of τ_s itself.

3.3 Directed reorganization

Early warning in the Systemic Tau framework is a statement about *change* in relational organization, not about the absolute level of τ_s . Given two regimes—a basal window B and an approach window A anchored to a candidate transition—define

$$\Delta\tau_s = \overline{\tau_s}^{(A)} - \overline{\tau_s}^{(B)}, \quad (8)$$

where bars denote temporal averages of $\tau_s(t)$ over the respective windows. The sign of $\Delta\tau_s$ carries structural meaning:

- $\Delta\tau_s > 0$: net *tightening* of cross-variable rank organization toward the event;
- $\Delta\tau_s < 0$: net *loosening* or inversion of that organization;
- $\Delta\tau_s \approx 0$: no detectable directed relational reorganization at the chosen scale.

Thus $\Delta\tau_s$ is a directed relational early-warning indicator. It does not presuppose that transitions always increase variance; it asks whether the pattern of mutual organization is changing, and in which direction Padilla-Villanueva [2025a, 2026a].

3.4 Relation to Lyapunov exponents and classical EWS

Lyapunov exponents quantify local expansion rates in state space and remain the gold standard for characterizing deterministic chaos when the equations of motion are known or high-quality reconstructions are available Wolf et al. [1985]. Systemic Tau is complementary rather than competitive: it is non-parametric, applicable to short and noisy series, and targeted at *ordinal coupling* rather than infinitesimal divergence. Relative to classical EWS, τ_s replaces the question “is the system fluctuating more?” with “is the mutual rank geometry reorganizing?” The two questions coincide only in special cases; in general they probe distinct layers of dynamical structure.

4 The RECD Framework: Hierarchical Levels of Ordinal Conjunctions

The Discrete Extramental Clock (RECD; *Reloj Extramental Discreto*) formalizes the idea that discrete system time need not be the uniform tick of an external chronometer. Instead, time advances when multivariate ordinal structure coheres at a sufficient depth. The foundational object of the present formulation is a hierarchy of *ordinal conjunctions* Φ_1, Φ_2, Φ_3 . A historically earlier construction—a Feigenbaum-type gate $g(\tau_s)$ that modulates chronological increments—is retained as a *regime cue* for the weights $\alpha_\ell(\lambda)$ (Section 5), not as a second, competing definition of the clock. Nested conjunctions

define *what* advances the RECD; Systemic Tau and Feigenbaum scaling define *how* the contributions of different depths are weighted.

4.1 Ordinal Pattern Representation

We encode each coordinate by Bandt–Pompe ordinal patterns Bandt and Pompe [2002], Amigó [2010]. Fix an embedding dimension $m \geq 2$ and a delay $\tau \geq 1$ (default operational choice: $m = 3, \tau = 1$). For coordinate i and time t , form the delay vector

$$\mathbf{v}_t^{(i)} = (X_t^{(i)}, X_{t+\tau}^{(i)}, \dots, X_{t+(m-1)\tau}^{(i)}) \quad (9)$$

and define the ordinal pattern $\pi_t^{(i)}$ as the unique permutation $\pi \in S_m$ that sorts $\mathbf{v}_t^{(i)}$ in nondecreasing order (with a fixed tie-breaking rule). The alphabet has cardinality $m!$; for $m = 3$ there are six admissible symbols. Collecting all coordinates yields the multivariate symbol process

$$S_t = (\pi_t^{(1)}, \dots, \pi_t^{(d)}) \in \{1, \dots, m!\}^d. \quad (10)$$

All subsequent RECD levels are functions of $\{S_t\}$ (and of short histories of S_t). The encoding is invariant under componentwise strictly increasing maps and focuses the analysis on relational geometry in symbol space rather than on absolute scale.

4.2 Three Hierarchical Levels of Conjunctions

Three planes (discipline). Throughout we keep three planes distinct and name them explicitly whenever any Level 3 quantity is discussed:

1. **Structural claims** — statements about the relational geometry of the joint symbol process S_t (e.g., $\text{Res}_{\text{pair}} > 0$).
2. **Estimators** — numerical readouts $\widehat{M}_\ell \in \{\Phi_1, \Phi_2, \text{excess3}, \Phi_3, \text{Res}_{\text{pair}}\}$ computed from finite windows.
3. **Aggregation policy** — the regime cue λ and schedule $\alpha_\ell(\lambda)$ that turn estimators into *aggregation shares* f_ℓ of ΔRECD .

Collapsing these planes is the main source of interpretive error (e.g., reading a rise in the aggregation share f_3 as proof of elevated irreducible residual when only α_3 has changed, or treating the proxy excess3 as if it were Res_{pair}).

Terminological contract. We use three families of terms and do not interchange them: *irreducible residual* (Res_{pair} , claim); *proxy* (excess3 , estimator); *aggregation share* (f_ℓ , weighted fraction under (λ, α)). An unqualified “Level 3” is always disambiguated by the surrounding sentence.

Level 1 — coincidence. The weakest conjunction is simultaneous equality of ordinal symbols across pairs of coordinates. With indicator $\mathbf{1}\{\cdot\}$,

$$\Phi_1(t) = \frac{2}{d(d-1)} \sum_{1 \leq i < j \leq d} \mathbf{1}\{\pi_t^{(i)} = \pi_t^{(j)}\} \in [0, 1]. \quad (11)$$

Level 1 is pure ordinal synchronization: a normalized coincidence rate. It is easy to estimate and closely related to mutual-information-type synchronization measures, but it does not by itself encode memory or higher-order structure. It is an order- ≤ 2 observable.

Level 2 — persistent relational structure. Level 2 does not require identical symbols. For each pair (i, j) define a discrete relation code

$$R_t^{ij} = \begin{cases} \text{EQ} & \text{if } \pi_t^{(i)} = \pi_t^{(j)}, \\ \text{GT} & \text{if } \pi_t^{(i)} > \pi_t^{(j)}, \\ \text{LT} & \text{if } \pi_t^{(i)} < \pi_t^{(j)}, \end{cases} \quad (12)$$

or, more generally, the ordered pair $(\pi_t^{(i)}, \pi_t^{(j)})$ as a joint relational state. Fix a persistence length $p_{\min} \geq 2$ (operational default $p_{\min} = 4$) and a dominance fraction $f_{\min} \in (0, 1]$ (default 0.75). Declare the pair persistent at time t if the same relation code occupies at least a fraction $f_{\min}$ of the retrospective window $\{t - p_{\min} + 1, \dots, t\}$. Then

$$\Phi_2(t) = \frac{2}{d(d-1)} \sum_{1 \leq i < j \leq d} w(R_t^{ij}) \mathbf{1}\{\text{pair } (i, j) \text{ is persistent at } t\}, \quad (13)$$

where $w(R)$ is a nonnegative weight (e.g., $w(\text{EQ}) = 1$ and $w(\text{GT}) = w(\text{LT}) = 0.85$). Level 2 adds memory and relational quality on the pattern graph McCullough et al. [2015], Padilla-Villanueva [2025b], still within pairwise (order- ≤ 2) structure.

Level 3 — irreducible residual (strong claim and proxy). Level 3 isolates joint ordinal configurations not reproducible by pairwise (order- ≤ 2) laws. Full comparison of claim and estimator is in Section 4.3; here we only fix notation.

Let $\mathcal{J}_t^{(w)}$ be the empirical joint law of S on a window of length w , and let $\mathcal{R}_{\text{pair}}$ be the class of pairwise interaction models (log-linear / exponential families with interactions of order ≤ 2). The **irreducible residual** (strong Level 3 claim) is the information-projection gap

$$\text{Res}_{\text{pair}}(t) := \inf_{Q \in \mathcal{R}_{\text{pair}}} D_{\text{KL}}(\mathcal{J}_t^{(w)} \| Q), \quad (14)$$

with D_{KL} in bits. When the information projection $P^{(2)}$ of $\mathcal{J}_t^{(w)}$ onto $\mathcal{R}_{\text{pair}}$ exists, $\text{Res}_{\text{pair}}(t) = D_{\text{KL}}(\mathcal{J}_t^{(w)} \| P^{(2)})$. Strong Level 3 holds when $\text{Res}_{\text{pair}}(t) > \theta_3$. In this sense Level 3 is not “more correlation”: correlation, τ_s , Φ_1 , and Φ_2 are order- ≤ 2 functionals, whereas Res_{pair} is the residual orthogonal (in the KL projection sense) to $\mathcal{R}_{\text{pair}}$. Strong Level 3 requires $d \geq 3$.

The continuous readout used throughout the synthetic sections is the hybrid *proxy* excess3 defined in Section 4.3; a binary activation

$$\Phi_3(t) = \mathbf{1}\{\text{excess3}(t) > \theta_3\} \quad (15)$$

is retained for event counting. Notation in the rest of the paper:

$$\widehat{M}_3 \in \{\text{excess3}, \Phi_3, \text{Res}_{\text{pair}}\}, \quad \Phi_3^* := \widehat{M}_3 \text{ in the weighted clock (31)}. \quad (16)$$

Rule of reading: “irreducible residual” always names the *claim* Res_{pair} ; numerical Level 3 *abundance* on the logistic suite always means the *proxy* excess_3 unless Res_{pair} is written explicitly. The two are not interchangeable.

Nesting of claims (not of binary indicators). Let Claim- ℓ be the structural assertion associated with level ℓ :

- Claim-1: simultaneous ordinal coincidence across coordinates (order- ≤ 2 synchronization);
- Claim-2: durable pairwise relational codes on a short history (order- ≤ 2 with memory);
- Claim-3: joint law not in the pairwise interaction class ($\text{Res}_{\text{pair}} > 0$).

Canonical nesting is a strict order of *logical strength of claim*, not of the default binary estimators:

$$\text{Claim-3} \succ \text{Claim-2} \succ \text{Claim-1}, \quad (17)$$

meaning: (i) Claim- ℓ asserts structure that Claim- $(\ell - 1)$ does not assert; (ii) truth of Claim- ℓ is not a consequence of Claim- k for any $k < \ell$; (iii) falsity of Claim-1 does not preclude Claim-2 in the default estimators (EQ is only one of several relational codes). The schematic inclusion $\Phi_3 \supset \Phi_2 \supset \Phi_1$ in earlier presentations should be read only in this claim sense. Operationally the estimators $\widehat{M}_\ell$ are computed *in parallel* from S_t and combined by weighted summation (1); one does *not* have $\Phi_3 = 1 \Rightarrow \Phi_2 = 1 \Rightarrow \Phi_1 = 1$ in general.

Hard event nesting can be *imposed by design* when the scientific question requires it—e.g., fire Level 3 only if $\text{Res}_{\text{pair}} > \theta_3$ and a Level 2 persistence gate holds—but that conjunction is an optional protocol, not a theorem about the free estimators. Soft nesting of claims (Type II) is always in force; hard nesting of indicators (Type I) is optional and must be declared.

4.3 Level 3: Irreducible Residual versus Operational Proxy

Three steps: weak vs. strong Level 3 (the latter is Res_{pair}); geometries where residual and proxy disagree; the hybrid proxy and its fixed coefficients.

Weak versus strong Level 3. Let $S_{[t-w+1, t]}$ denote a local window of multivariate symbols of length w , with empirical joint $\mathcal{J}_t^{(w)}$. Write $H(\cdot)$ for Shannon entropy (in bits) and $I(\pi^{(i)}; \pi^{(j)})$ for pairwise mutual information. The *total correlation* of the joint pattern is

$$\text{TC}(t) = \sum_{i=1}^d H(\pi^{(i)}) - H(\pi^{(1)}, \dots, \pi^{(d)}) \quad (18)$$

estimated on the local window. With P_{ind} the product of the empirical marginals, the standard identity

$$\text{TC}(t) = D_{\text{KL}}(\mathcal{J}_t^{(w)} \parallel P_{\text{ind}}) \quad (19)$$

identifies TC with the residual to *total independence* (weak Level 3: non-factorization). When the pairwise projection $P^{(2)}$ exists, TC decomposes as

$$\text{TC}(t) = \text{Res}_{\text{pair}}(t) + D_{\text{KL}}(P^{(2)} \parallel P_{\text{ind}}), \quad (20)$$

i.e., the irreducible residual beyond pairs plus multiinformation already explainable by pairwise structure. Weak Level 3 is $\text{TC} > 0$; strong Level 3 is $\text{Res}_{\text{pair}} > 0$. Only the latter justifies the phrase “irreducible residual beyond pairwise models.”

Where the irreducible residual and the proxy part ways. Two canonical geometries make the distinction operational rather than verbal:

- **Common-driver.** Three symbols driven by a shared latent factor with no genuine triple interaction. Pairwise mutual informations are large; the joint is essentially in $\mathcal{R}_{\text{pair}}$. Then $\text{Res}_{\text{pair}} \approx 0$, while TC, Surp, and often the proxy excess3 are *elevated*. Reading excess3 as the irreducible residual is false here.
- **XOR-type (higher-order joint).** A third symbol is (near) deterministic function of two others with vanishing pairwise marginal dependence. Then pairwise MI is small, $\text{Res}_{\text{pair}} > 0$, and Syn (hence excess3) typically rises. Irreducible residual and proxy agree directionally.

Geometry	TC / Surp	proxy excess3	Res_{pair}
Common-driver	high	often high	≈ 0
XOR-type joint	moderate-high	high	> 0
Near independence	≈ 0	≈ 0	≈ 0

Thus excess3 is a useful abundance probe and a weak-Level 3 surrogate; it is not a certified irreducible residual. Any claim that “Level 3 has risen” must state which of $\{\text{TC}, \text{excess3}, \text{Res}_{\text{pair}}\}$ is meant.

The hybrid proxy excess3. A scalable proxy uses a pairwise baseline

$$I_{\text{pair}}(t) = (d-1) \frac{2}{d(d-1)} \sum_{i < j} I(\pi^{(i)}; \pi^{(j)}), \quad (21)$$

i.e., a scaled average of pairwise mutual informations (the factor $d-1$ is a methodological contract making the baseline dimensionally comparable to a multiinformation residual). The excess of TC beyond this baseline is

$$\text{Syn}(t) = [\text{TC}(t) - I_{\text{pair}}(t)]_+. \quad (22)$$

Syn is a heuristic residual-like term, *not* the exact I -projection gap (14): it subtracts a pairwise summary rather than projecting onto $\mathcal{R}_{\text{pair}}$. A second component measures the surprise of observed joint tuples under an independence model. If $p_{\text{obs}}(s)$ is the empirical frequency of joint symbol s and $p_{\text{ind}}(s) = \prod_{i=1}^d p_i(s_i)$ the product of marginals, define the weighted positive log-ratio

$$\text{Surp}(t) = \sum_s p_{\text{obs}}(s) [\log_2 p_{\text{obs}}(s) - \log_2 p_{\text{ind}}(s)]_+. \quad (23)$$

One has $\text{Surp} \geq \text{TC}$, with equality only when the information density does not change sign on the support of p_{obs} ; Surp alone does not separate pairwise from higher-order structure.

The continuous Level 3 readout used throughout is the hybrid

$$\text{excess3}(t) = 0.6 \text{Syn}(t) + 0.4 \text{Surp}(t). \quad (24)$$

The pair $(0.6, 0.4)$ is a declared a priori design choice, not a fit to the logistic suite Padilla-Villanueva [2025b]: Syn is overweighted because it is residual-like (Surp alone fires under common-driver pairwise dependence); the weights are frozen for falsifiability; and on the ensemble of Section 7 the same directional contrasts appear in pure Syn and pure Surp , so mild reweighting (e.g. $(0.7, 0.3)$ or $(0.5, 0.5)$) does not flip the ordinal conclusions. Scientific claims in this paper are therefore about the hybrid *class*, not about a unique coefficient pair.

This construction is a computable **proxy** for higher-order joint structure in the broad sense of partial information decomposition Williams and Beer [2010], Lizier [2014]; it does not claim a unique axiomatic PID atom and is *not* Res_{pair} . In particular:

$$\text{excess3} > 0 \not\Rightarrow \text{Res}_{\text{pair}} > 0, \quad (25)$$

because Surp can fire on pure pairwise dependence (common-driver). Honest reporting should therefore separate $(\text{Syn}, \text{Surp}, \text{excess3})$ and, when d and alphabet size permit, estimate Res_{pair} directly (e.g., iterative proportional fitting to pairwise margins plus KL).

Clock algebra. The unweighted RECD increment at the level of raw conjunctions is

$$\Delta \text{RECD}_0(t) = \Phi_1(t) + \Phi_2(t) + \widehat{M}_3(t), \quad (26)$$

with default $\widehat{M}_3 = \text{excess3}$ (proxy; or Φ_3 for binary counting). Weighted RECD replaces unit coefficients by $\alpha_\ell(\lambda)$. When a Res_{pair} estimator is available it may replace excess3 as $\widehat{M}_3$ without changing the clock algebra; the structural interpretation of the Level 3 term then upgrades from proxy to irreducible residual.

5 Weighted RECD and Regime-Dependent Aggregation

The schedule $\alpha_\ell(\lambda)$ is the most easily over-read object in the framework. This section states what it is—and what it is not. In short: it lives entirely on the aggregation plane of Section 4.2; it does not create geometry on the claim plane.

5.1 Status of the aggregation schedule: policy, not geometry

We separate three objects that must not be collapsed:

1. $\lambda(t) \geq 0$ — a dimensionless *regime cue*;
2. $\alpha_\ell(\lambda)$ — a *weight schedule* for level ℓ ;
3. $f_\ell(t)$ — the resulting *aggregation share* of ΔRECD (defined below).

Neither λ nor α is an observable of the joint symbol law (unlike Res_{pair} or excess3). They implement a regime-aware aggregation policy: a map from estimators $\widehat{M}_\ell$ to a composite increment, chosen under admissibility and then held fixed. (Older notes used “ontological weight” for what we call the aggregation share f_ℓ below.)

Admissible weighting engine. A pair (λ, α) is *admissible* if

1. $\alpha_\ell(\lambda) > 0$ for all ℓ (no level is hard-killed);
2. *depth monotonicity*: as λ increases in the regime of interest, $\alpha_1 \downarrow$, $\alpha_2 \uparrow$ or stabilizes, $\alpha_3 \uparrow$;
3. λ is not dominated by a circular function of $\widehat{M}_3$ alone;
4. the form of α and the construction of λ are pre-specified (or calibrated on a labeled holdout separate from the Level 3 test);
5. λ satisfies the design goal

$$\mathbb{E}[\lambda \mid \mathcal{R}_{\text{chaos}}] > \mathbb{E}[\lambda \mid \mathcal{R}_{\text{pre}}] \quad (27)$$

on the system of interest (Appendix A).

If no reliable cue exists, report unweighted RECD ($\alpha_\ell \equiv 1$).

Reference cue. A reference empirical cue couples Systemic Tau to the chaotic band of (7):

$$\lambda_\tau(t) = \max\left(0, s \frac{|\tau_s(t)| - \tau_{\text{ch}}}{\tau_{\text{ch}}}\right) \cdot (1 + \gamma \log \delta), \quad (28)$$

where $\tau_{\text{ch}} = 0.41$, $\delta \approx 4.6692016091$ is Feigenbaum’s constant used here only as a *dimensionless scale prefactor* (role δ_{cue} ; see Section 6), $\gamma \geq 0$ is a mild prefactor (default $\gamma = 0.2$), and $s \in \{+1, -1\}$ is a system-calibrated *polarity* (Appendix A). When $s = +1$, λ_τ grows as $|\tau_s|$ exceeds τ_{ch} ; when $s = -1$, it grows as $|\tau_s|$ falls into the chaotic band $|\tau_s| < \tau_{\text{ch}}$. On many desynchronizing transitions one expects $s = -1$; the logistic ensemble of Section 7 has the opposite ordering and needs polarity recalibration or multi-cue fallback (Appendix A). Polarity is part of the cue’s formal status. With known control parameter r , replace (28) by a monotone map $r \mapsto \lambda(r)$ to isolate regime effects from cue noise.

Reference schedule and weighted clock. One admissible reference schedule (not the unique map compatible with depth monotonicity) is

$$\begin{aligned} \alpha_1(\lambda) &= \alpha_{10} e^{-\beta_1 \lambda}, \\ \alpha_2(\lambda) &= \alpha_{20} (1 + \gamma_2 \lambda), \\ \alpha_3(\lambda) &= \alpha_{30} (1 + \gamma_3 \lambda + \delta_3 \lambda^2), \end{aligned} \quad (29)$$

with baseline coefficients $\alpha_{\ell 0} > 0$ and shape parameters $\beta_1, \gamma_2, \gamma_3, \delta_3 \geq 0$ (reference defaults: $\alpha_{\ell 0} = 1$, $\beta_1 = 2$, $\gamma_2 = 1.5$, $\gamma_3 = 3$, $\delta_3 = 2$; here δ_3 is a shape coefficient, *not* Feigenbaum’s δ). Consequently, as λ increases,

$$\alpha_1(\lambda) \downarrow, \quad \alpha_2(\lambda) \uparrow, \quad \alpha_3(\lambda) \uparrow\uparrow. \quad (30)$$

The weighted RECD update is

$$\Delta\text{RECD}(t) = \alpha_1(\lambda) \Phi_1(t) + \alpha_2(\lambda) \Phi_2(t) + \alpha_3(\lambda) \Phi_3^*(t), \quad (31)$$

where $\Phi_3^* = \widehat{M}_3$ is either binary Φ_3 , continuous proxy excess3, or irreducible residual Res_{pair} when estimated (16). Accumulated extramental time is the partial sum

$$T_{\text{RECD}}(t) = \sum_{u \leq t} \Delta\text{RECD}(u). \quad (32)$$

The schedule α is an axiomatized design object together with a pre-specified representative family. It is not a natural law derived from $\mathcal{J}_t^{(w)}$ alone, and it does not create Level 3 geometry where none exists in $\widehat{M}_3$.

Relation to an earlier gating construction. An earlier formulation advanced chronological increments through a gate $g(\tau_s)$ with Feigenbaum renormalization δ^{-k} in chaotic runs Padilla-Villanueva [2026b]. In the hierarchy developed here that gate is absorbed into λ and $\alpha_\ell(\lambda)$: it modulates the *aggregation share* of conjunction depths rather than defining a parallel clock. There is therefore a single RECD object—nested ordinal conjunctions—with Systemic Tau and Feigenbaum-motivated scales as its weighting engine. Feigenbaum’s δ appears in two historically related but conceptually distinct roles—cue scale factor (δ_{cue} in (28)) versus legacy time-step reduction (δ^{-k} in the earlier gate)—which must not be conflated with the shape coefficient δ_3 in (29).

5.2 Aggregation shares across regimes

The aggregation share of level ℓ is

$$f_\ell(t) = \frac{\alpha_\ell(\lambda) \Phi_\ell^*(t)}{\sum_{k=1}^3 \alpha_k(\lambda) \Phi_k^*(t)}, \quad (33)$$

whenever the denominator is positive. Level 1 events are abundant but thin in claim; Level 2 carries relational memory; Level 3, as residual or declared proxy, carries joint structure pairwise models do not fully explain. Admissible α encodes the design hypothesis that near chaos the composition of the clock increment shifts toward deeper layers (f_3 rises relative to f_1)—a hypothesis about aggregation, not a theorem about $\text{Res}_{\text{pair}}(r)$.

Abundance versus aggregation share (mandatory distinction). Two Level 3 quantities must never be identified:

1. **Abundance** — raw proxy excess3, irreducible residual Res_{pair} , or high-Level 3 rates. These live on the estimator plane and do *not* require λ or α .
2. **Aggregation share** f_3 — a regime-aware construct on the aggregation plane whose validity inherits from the quality of λ .

A rise in f_3 with flat abundance can be pure reweighting. Claims about Level 3 structure should lead with abundance; f_3 is reported only with the declared triple $(\widehat{M}_\ell, \lambda, \alpha)$.

Section 7 tests that proxy abundance (excess3) rises from pre-chaotic to chaotic regimes, and—under ground-truth $\lambda(r)$ —that f_3 rises as well. Only the former is a pure geometric claim about $\widehat{M}_3$.

6 Relation to Chaos Theory and the Feigenbaum Route

The period-doubling route to chaos supplies the dynamical backdrop for the operational bands and cue scales above Feigenbaum [1978, 1983]. For the logistic family

$$x_{n+1} = rx_n(1 - x_n), \quad r \in [0, 4], \quad (34)$$

an infinite cascade of period-doubling bifurcations accumulates at

$$r_\infty \approx 3.56995, \quad (35)$$

with successive bifurcation spacings asymptotically governed by Feigenbaum’s constant

$$\delta = \lim_{n \rightarrow \infty} \frac{r_n - r_{n-1}}{r_{n+1} - r_n} \approx 4.6692016091. \quad (36)$$

Beyond r_∞ the attractor is chaotic on a set of positive measure in r , interrupted by periodic windows.

What is inherited, and what is analogy. We distinguish carefully:

- **Inherited from Feigenbaum universality:** the existence of a universal accumulation point r_∞ and a universal scaling constant δ for unimodal maps of the period-doubling class; the availability of a dimensionless geometric scale independent of laboratory units.
- **Operational analogy (not derivation):** the numerical bands $\tau_s \geq 0.50$, $|\tau_s| < 0.41$, $\tau_s \leq -0.41$; the use of δ as a mild prefactor in λ_τ ; the prediction that Level 3 *abundance* is elevated on the chaotic side of the cascade relative to low-period stations. None of these is a theorem deduced from Feigenbaum’s functional equation; they are calibrated, testable design choices motivated by the cascade geometry.

Notation for δ : in (28) the symbol denotes Feigenbaum’s constant (role δ_{cue}); the coefficient δ_3 in (29) is unrelated. Legacy constructions that used δ^{-k} as a time-step gate Padilla-Villanueva [2026b] are a third, historical role and are not used as the primary clock here.

Why the cascade is still useful for an ordinal clock. Three operational reasons remain after that sobriety check.

Universal scales without dataset-specific tuning. The constant δ and the geometric contraction of parameter intervals supply dimensionless scales independent of the laboratory units of a time series. Importing δ into λ as a mild prefactor is a way of tying cue magnitude to a known universal structure rather than to *ad hoc* percentiles of a single dataset Padilla-Villanueva [2026b]—without claiming that λ itself is universal.

Ordinal collapse near the chaotic band. As r approaches and exceeds r_∞ , local predictability collapses and rank relationships among coupled coordinates become highly volatile. The operational band $|\tau_s| < 0.41$ is *calibrated* to mark this intermediate regime of ordinal instability—neither frozen synchronization nor trivial independence—where relational reorganization is expected to be most informative for early warning. The number 0.41 is not derived from δ ; it is a Feigenbaum-motivated operational threshold.

Depth-dependent abundance along the cascade. In pre-chaotic periodic regimes, multivariate ordinal patterns are often dominated by coincidence and short relational codes: Levels 1–2 often suffice. In developed chaos, pairwise descriptions can become incomplete; joint configurations rare under independence may appear more frequently, elevating Syn and Surp and hence excess3. Weighted RECD can *amplify* that geometric effect by increasing $\alpha_3(\lambda)$ when the regime cue indicates chaos, but amplification is aggregation, not geometry. The testable predictions are therefore:

1. raw Level 3 *abundance* (proxy excess3; and, with adequate window, the irreducible residual Res_{pair}) is regime-sensitive along the cascade—a claim about joint symbol geometry, independent of α ;
2. the aggregation share f_3 of ΔRECD rises when an admissible $\alpha_3(\lambda)$ tracks the regime—a claim about the aggregation policy.

Both are examined in Section 7; only (i) is a pure Level 3 geometric claim about estimators. Empirically, on the dense logistic sweep the proxy excess3 peaks near accumulation rather than in deep chaos (Section 7.4). That peak location is a synthetic *observation*; the geometric account we attach to it is a *working hypothesis*, not a result of the present paper.

The bands $\tau_s \geq 0.50$ and $\tau_s \leq -0.41$ are ordinal analogues of strong synchronization and strong anti-phase organization Pecora and Carroll [1990], Mainieri and Rehacek [1999]; the chaotic band interpolates between them for early-warning design.

7 Synthetic Validation on Coupled Logistic Maps

We summarize a controlled synthetic experiment that tests a central dynamical prediction of the nested hierarchy: that chaotic regimes elevate Level 3 *abundance* (here: the proxy excess3) relative to pre-chaotic regimes, and that—under ground-truth $\lambda(r)$ —the aggregation share f_3 rises as well Padilla-Villanueva [2025b], Kaneko [1992].

7.1 Protocol

Consider $d = 4$ diffusively coupled logistic maps with weak coupling $\varepsilon = 0.05$ and small observational noise $\sigma = 0.003$. Two regimes are contrasted with fixed ground-truth control parameters:

- **Pre-chaotic:** $r = 3.30$ (period-doubling cascade, pre-accumulation);
- **Chaotic:** $r = 3.85$ (post-Feigenbaum developed chaos).

For each regime we generate $N_{\text{real}} = 25$ independent realizations of length $T = 3200$ (after discarding a short transient). Ordinal patterns use $m = 3$, delay 1; Level 2 persistence uses $p_{\min} = 4$ and $f_{\min} = 0.75$; Level 3 uses window $w = 13$ and the hybrid excess3 of (24). Primary raw metrics (independent of α) are

Table 1: Regime-dependent Level 3 abundance (excess3, highL3) and aggregation share f_3 on coupled logistic maps ($d = 4$, $\varepsilon = 0.05$, $N_{\text{real}} = 25$). Values are mean $\pm$ std across realizations. Share f_3 uses ground-truth $\lambda(r)$, not $|\tau_s|$.

Metric	Pre ($r = 3.30$)	Chaos ($r = 3.85$)	Relative change
mean excess3 (abundance)	1.760 ± 0.004	1.892 ± 0.006	+7.5%
highL3 rate (> 1.75)	0.502 ± 0.020	0.904 ± 0.024	+80% (relative)
share f_3 (aggregation)	0.613	0.870	+0.257 abs.

- mean excess3;
- high-Level 3 rate: $\mathbb{P}(\text{excess3} > 1.75)$.

Weighted contributions use ground-truth regime labels for λ (equivalently $\alpha_3 = 1$ pre-chaos and $\alpha_3 = 6$ in chaos) so that the aggregation-share hypothesis can be tested without confounding by proxy inversion of $|\tau_s|$ on this particular coupled system.

7.2 Results

Table 1 reports the principal contrasts across $N_{\text{real}} = 25$ realizations. Figures 1–3 illustrate representative single-trajectory structure under the same protocol (ground-truth regime weights for α).

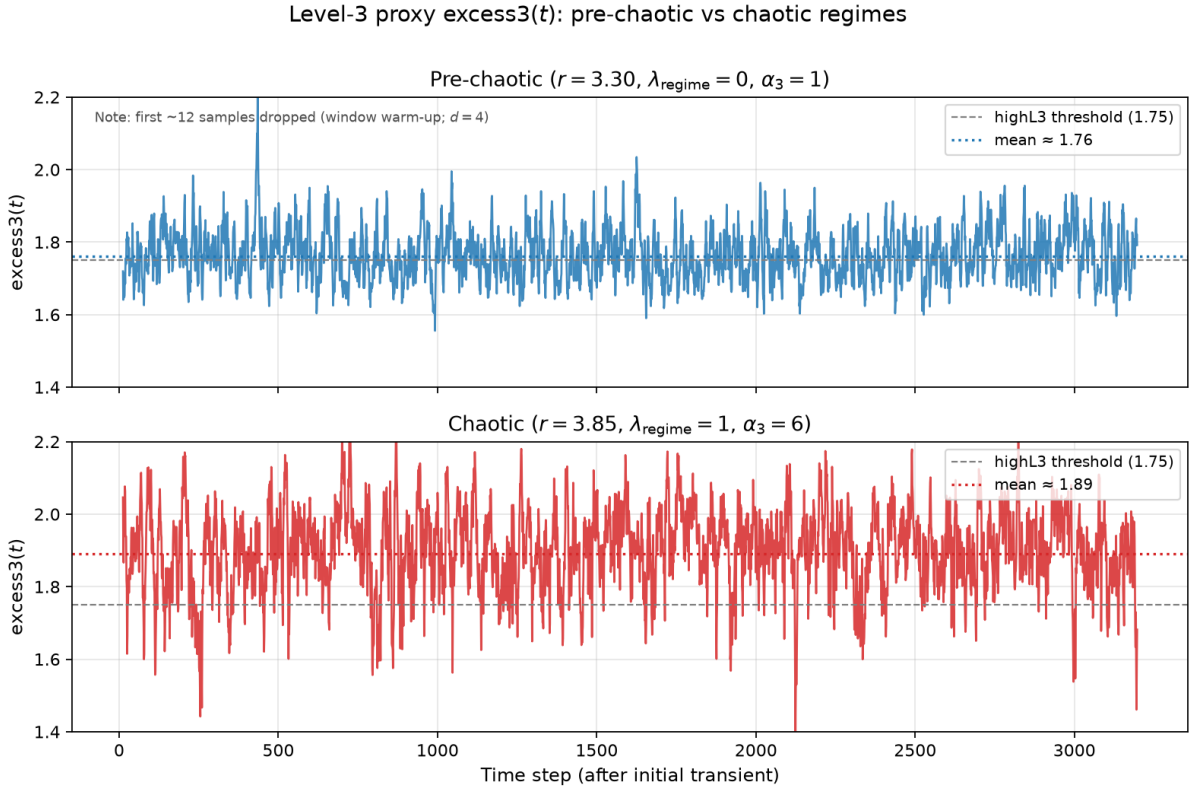


Figure 1: Level 3 proxy $\text{excess3}(t)$ on representative realizations of diffusively coupled logistic maps ($d = 4$, $\varepsilon = 0.05$). Top: pre-chaotic ($r = 3.30$). Bottom: chaotic ($r = 3.85$). Dashed line: highL3 threshold 1.75; dotted line: trajectory mean. Mean excess3 rises from ≈ 1.76 to ≈ 1.89 across the ensemble (Table 1). This is abundance of the proxy, not Res_{pair} .

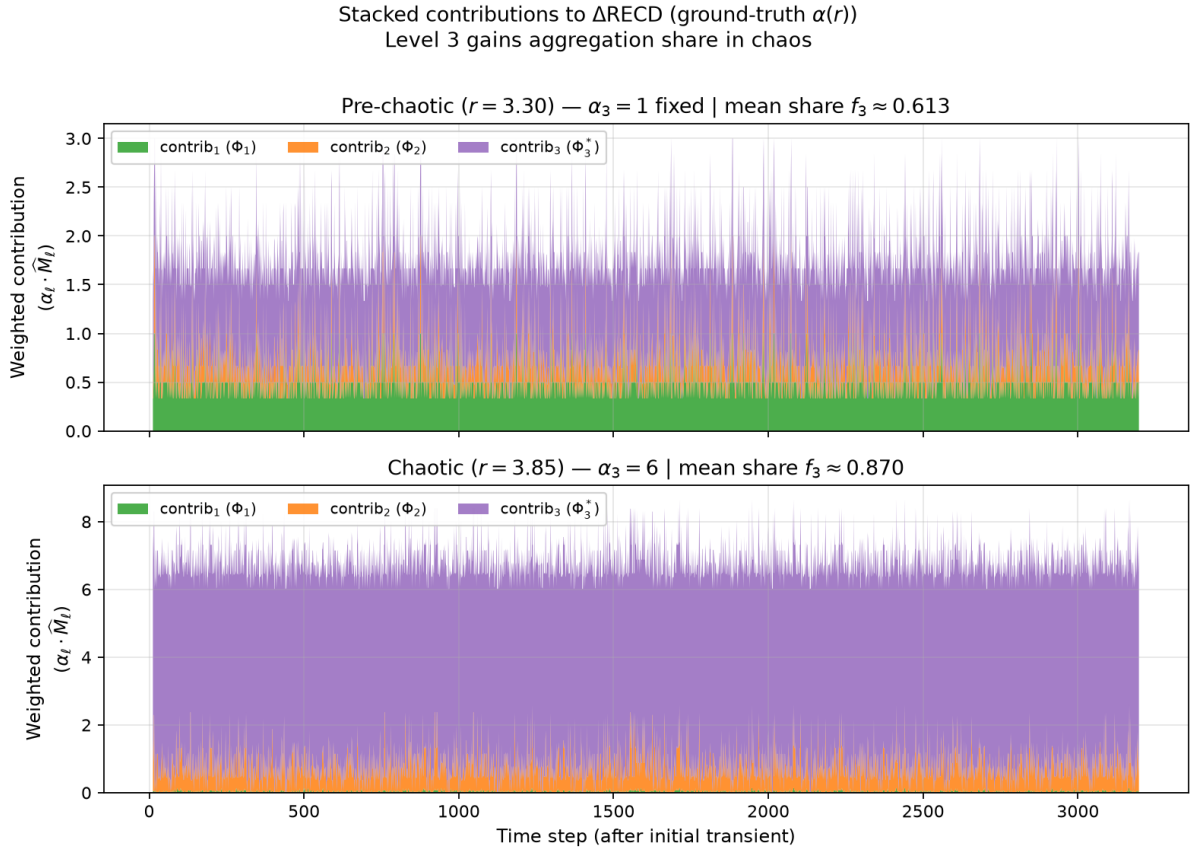


Figure 2: Stacked weighted contributions $\alpha_\ell \Phi_\ell$ to ΔRECD under ground-truth regime labels ($\alpha_3 = 1$ pre-chaos; $\alpha_3 = 6$ chaos). Level 3 (top band) gains aggregation share: mean f_3 increases from ≈ 0.613 to ≈ 0.870 .

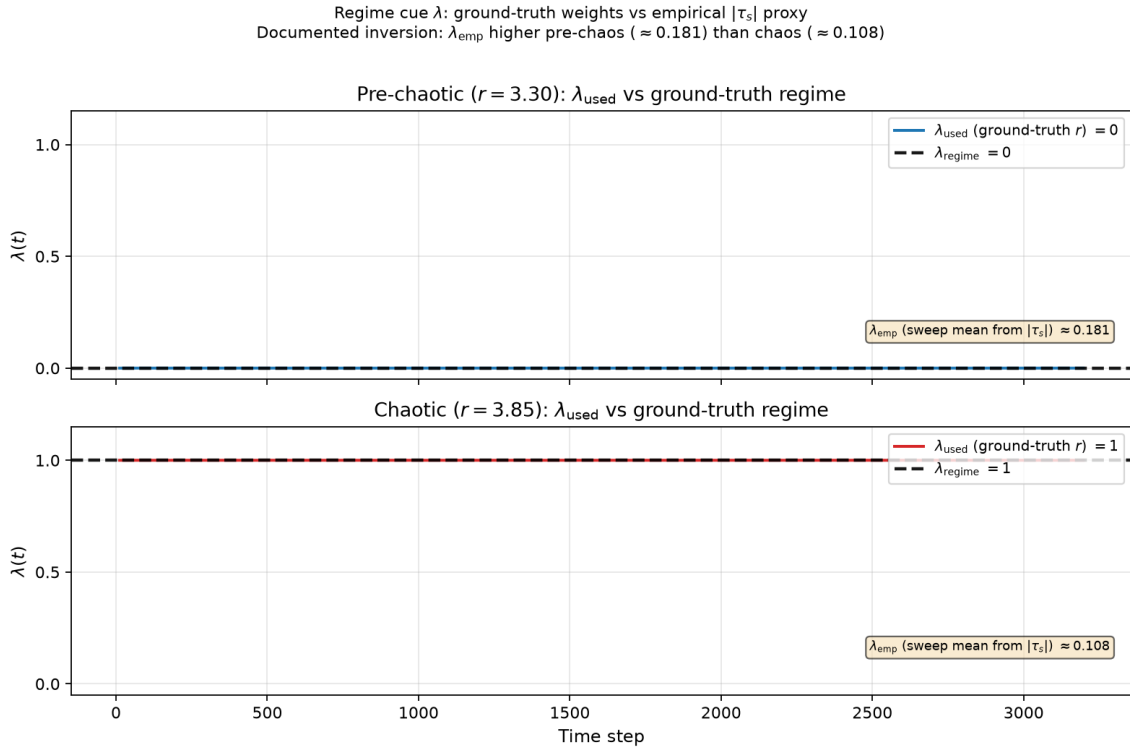


Figure 3: Regime cue λ : solid curves show λ_{used} from ground-truth r (the weights actually applied in Figures 1–2); dashed lines mark $\lambda_{\text{regime}} \in \{0, 1\}$. Annotated λ_{emp} values are sweep means of the pure $|\tau_s|$ -derived proxy, which is *higher* pre-chaos (≈ 0.181) than in chaos (≈ 0.108) on this system—the documented polarity inversion motivating Appendix A.

Raw Level 3 abundance (excess3) increases consistently: mean excess3 rises by approximately 7.5% (Figure 1), and the fraction of time spent above the high threshold nearly doubles. Under intentional reweighting by ground-truth regime, the aggregation share of ΔRECD attributable to Level 3 increases from roughly 61% to 87% (Figure 2). Concurrently, mean Φ_1 and Φ_2 tend to decrease slightly in chaos, as expected when simple coincidence and clean persistent codes become rarer under mixing—reinforcing that the excess3 rise is not a trivial rescaling of Level 1.

Light phase-shuffle surrogates that destroy cross-variable dependence while preserving marginal spectra Theiler et al. [1992] confirm that, in the pre-chaotic regime, observed excess3 lies significantly above the independence null. In chaos the absolute baseline of the excess metric shifts, so absolute p -values against surrogates must be interpreted jointly with the pre $\rightarrow$ chaos *delta*, which remains robust. A secondary methodological finding, highlighted in Figure 3, is that empirical $|\tau_s|$ on this coupled logistic proxy can be *higher* in the pre-chaotic synchronized regime than in developed chaos; hence $\lambda(|\tau_s|)$ requires system-specific calibration when ground-truth r is unavailable (Appendix A). This does not undermine excess3 as a raw Level 3 observable; it disciplines the use of τ_s -derived weights.

7.3 Interpretation (binary contrast)

Within the limits of this synthetic protocol, the data support a directional hierarchical prediction for the proxy abundance excess3 and, under ground-truth $\lambda(r)$, for the aggregation share f_3 . They do not by themselves establish that Res_{pair} tracks excess3 at short windows (Section 7.4). The RECD is therefore not a homogeneous counter of coincidences; under a declared engine its composition tracks both active structure depth and the chosen weights.

7.4 Dense cascade sweep

The two-point contrast rejects the coarse null that excess3 abundance is regime-invariant, but does not locate where along the period-doubling cascade abundance peaks, nor how it relates to classical EWS or to the irreducible residual Res_{pair} . We therefore recompute the nested pipeline on a dense grid of 32 values $r \in [3.10, 3.95]$ (densified near $r_\infty \approx 3.56995$), with $N_{\text{real}} = 6$ independent trajectories of length $T = 2200$ after burn-in, and the same $(d, \varepsilon, \sigma, m, w)$ as above. Reported metrics are mean excess3, highL3 rate, mean Res_{pair} (IPF pairwise maxent, stride 8), mean $|\tau_s|$, and classical variance and lag-1 autocorrelation (channel means).

Principal numerical findings on this ensemble are as follows (synthetic evidence, not analytic theorems):

1. **Coarse mono for the proxy.** At the reference stations $r = 3.30 \rightarrow 3.85$, mean excess3 rises ($1.76 \rightarrow 1.86$) and highL3 nearly doubles ($0.47 \rightarrow 0.84$), matching Table 1 directionally under the denser protocol.
2. **Peak location of the proxy.** *Observation (result of the suite).* Maximum mean excess3 on the grid is near pre-accumulation ($r \approx 3.566$, $\text{excess3} \approx 2.38$), not at developed chaos. Developed chaos retains elevated highL3 rates but intermediate mean excess3. This is a robust synthetic fact on the present ensemble. *Working hypothesis (not a result).* We tentatively interpret the peak as few-state high-period joints near r_∞ (non-product support elevating Surp and, when pairwise summaries under-explain TC, Syn) versus broader mixing in deep chaos (which can dilute the hybrid scores even when Res_{pair} at

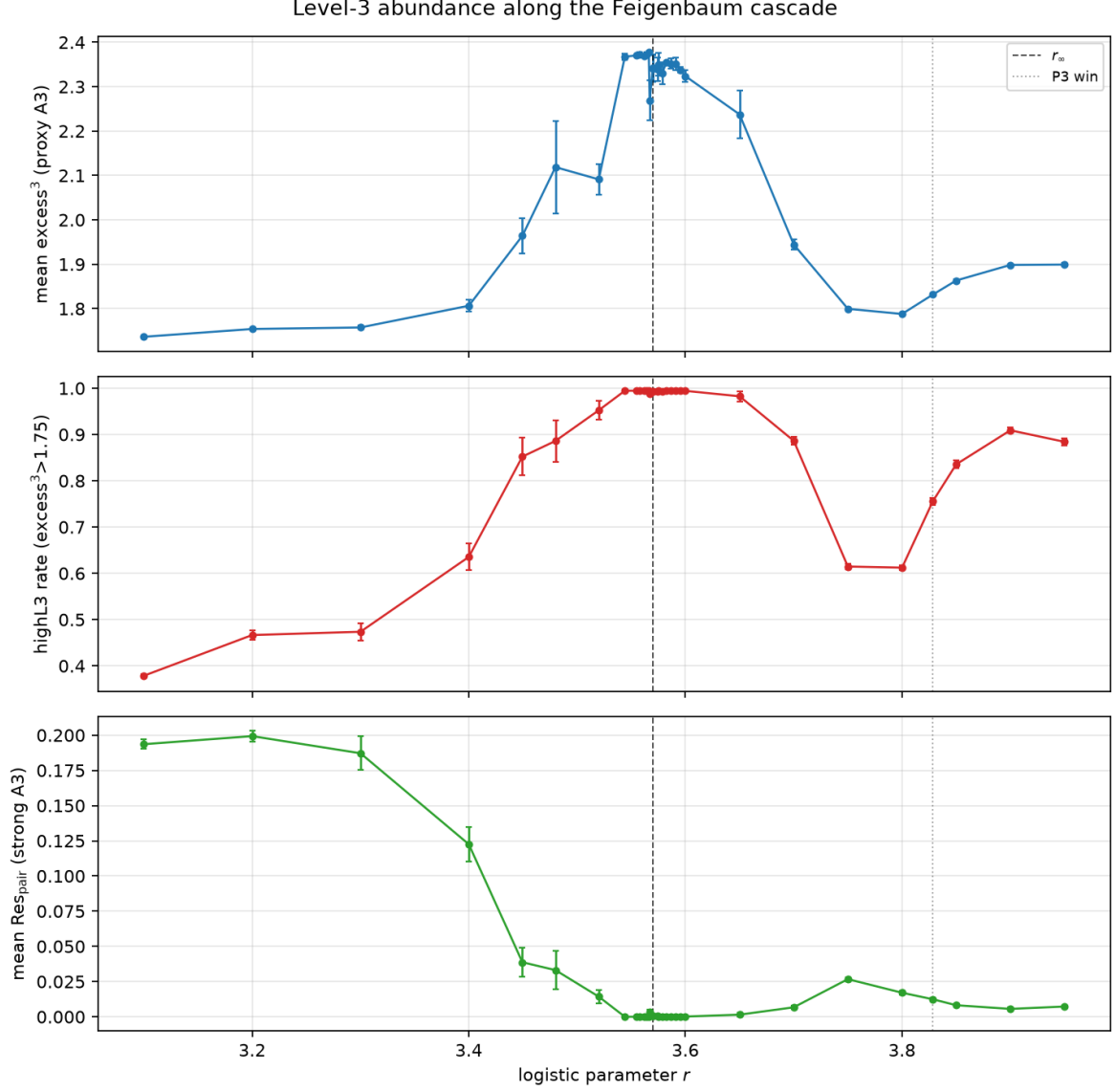


Figure 4: Level 3 abundance along the Feigenbaum cascade (coupled logistic maps, $d = 4$, $\varepsilon = 0.05$). Top: mean excess₃ (proxy A_3). Middle: highL3 rate. Bottom: mean Res_{pair} (irreducible residual). Vertical dashed line: r_∞ ; dotted: period-3 window onset. Error bars: SEM across realizations. excess₃/highL3 rise from period-2 stations into the accumulation band; the proxy maximum lies near r_∞ , not in deep chaos. The bottom panel uses the dense-sweep window $w = 13$; under that protocol Res_{pair} appears highest in low-period stations (Fig. 6 shows the window-length correction). The proxy and the irreducible residual are not interchangeable.

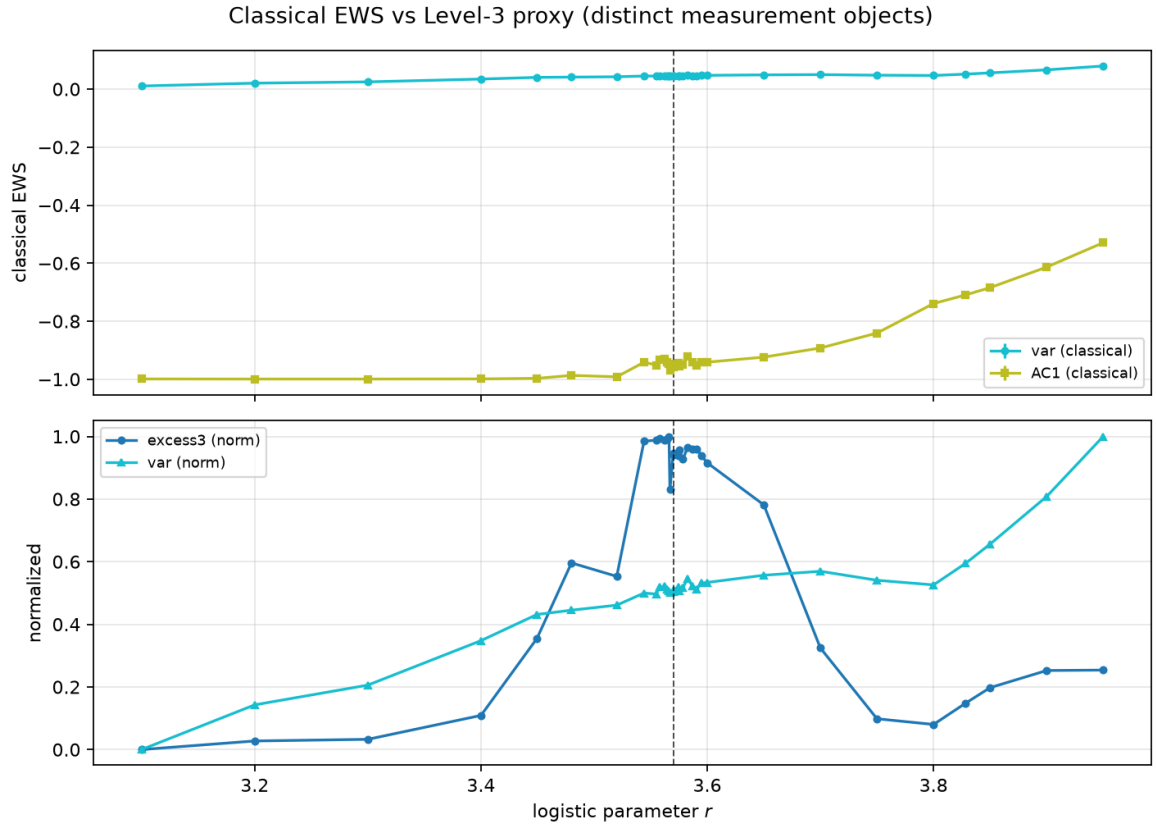


Figure 5: Classical EWS versus Level 3 proxy along the same r -grid. Top: channel-mean variance and lag-1 autocorrelation. Bottom: shapes of mean excess3 and variance after separate min-max normalization. Normalized variance does not reproduce the excess3 profile ($\text{corr}_r \approx 0.21$), consistent with the separation of measurement objects in Section 6.

adequate window length remains large). The window diagnosis below is *consistent with* this hypothesis but does not establish it. A controlled limit of $(\text{Syn}, \text{Surp}, \text{Res}_{\text{pair}})$ along the cascade remains future work.

3. **Irreducible residual and window length.** Under the dense-sweep protocol ($w = 13$), Res_{pair} appears largest in period-2 stations (≈ 0.20) and near zero in chaos. A window-length, alphabet-support, and IPF diagnostic shows that this is largely an estimator effect: at $r = 3.85$, mean Res_{pair} rises from ≈ 0.009 ($w = 13$) to ≈ 1.26 ($w = 104$) and ≈ 0.45 on a long contiguous block, while the IPF procedure converges in all stations. Near r_∞ the residual remains low even at long w (few-state high-period joint, nearly pairwise-explainable). With adequate w , the coarse ordering $\text{Res}(\text{chaos}) > \text{Res}(\text{period-2})$ is restored. Claims of the irreducible residual should still separate the proxy excess3 from Res_{pair} and report the window length.
4. **Non-triviality vs $|\tau_s|$ and variance.** Across r , $\text{corr}(\text{excess3}, |\tau_s|) \approx 0.27$ and $\text{corr}(\text{excess3}, \text{var}) \approx 0.21$. Systemic Tau and nested RECD answer dual questions; classical EWS answer a third.

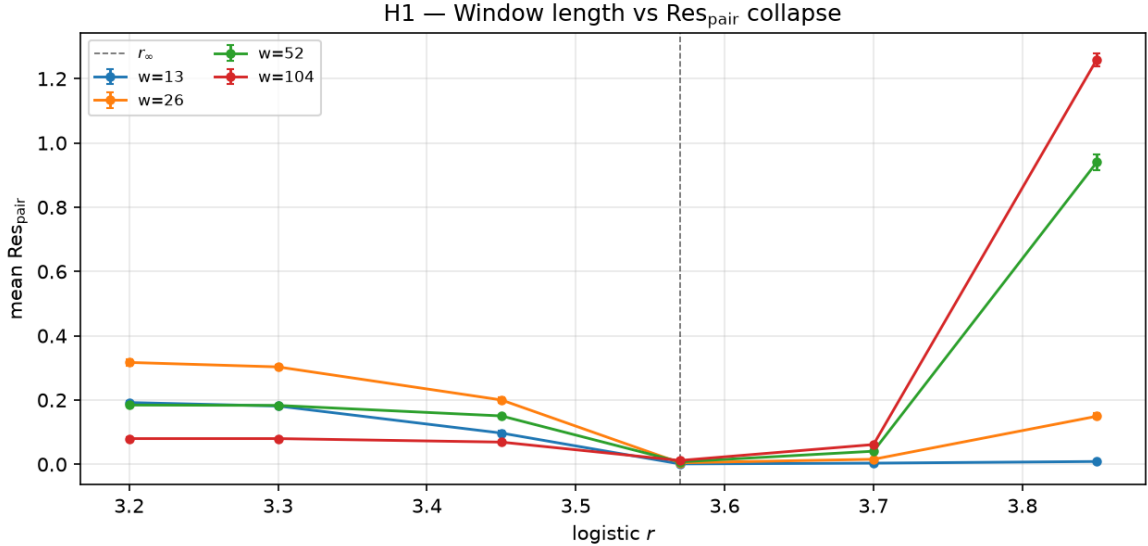


Figure 6: Window-length diagnosis of Res_{pair} at cascade stations. Short windows ($w = 13$) suppress the residual in developed chaos; long windows restore a large residual there, while near r_∞ the residual remains close to zero. The elevated period-2 value at $w = 13$ is largely short-window inflation relative to the long-block estimate.

Full numerical tables, companion panels (variability of excess3 and $|\tau_s|$; $\Phi_1/\Phi_2/f_3$; residual diagnostics), and reproducible scripts are provided with the supplemental materials. The algorithmic implementation is the open `nested-recd` library Padilla-Villanueva [2025b].

8 Discussion

8.1 Conceptual implications

The framework advanced here casts early warning as a problem of *hierarchical ordinal structure* rather than of univariate fluctuation size alone. Systemic Tau supplies a directed scalar of relational reorgani-

zation; nested RECD supplies a depth-resolved, event-driven clock. Together they answer two distinct but complementary questions: whether mutual organization is changing, and at what ordinal depth that change is expressed—provided one keeps structural claims, estimators, and aggregation weights on separate planes.

First, the sign of $\Delta\tau_s$ is scientifically meaningful rather than a nuisance: tightening and loosening are different reorganizations, and both may precede critical transitions depending on the substrate. Second, the RECD hierarchy makes explicit that not all ordinal events are equal: coincidence, persistence, and the irreducible residual Res_{pair} occupy different layers of structural claim, and excess3 is only a proxy for the third. Third, Feigenbaum-motivated scales enter λ and the operational bands as calibrated design structure, not as a derivation of Res_{pair} from Feigenbaum’s functional equation.

8.2 Advantages relative to classical EWS

Compared with variance and lag-1 autocorrelation, Systemic Tau and RECD are multivariate by construction, monotone-invariant, and targeted at coupling geometry. They do not assume a local fixed-point bifurcation, and they produce a signed relational contrast rather than a one-sided “rising indicator” narrative. The Level 3 layer—preferably Res_{pair} when estimable, otherwise excess3 with explicit caveats—is designed to remain informative when pairwise synchronization measures saturate or reverse.

8.3 Relation to neighbouring methodologies

The framework sits in a neighbourhood of existing ordinal and information-theoretic tools, but is not reducible to any one of them.

- **Permutation entropy and ordinal networks.** Bandt–Pompe encoding and ordinal networks Bandt and Pompe [2002], Amigó [2010], McCullough et al. [2015] provide the symbolic substrate of RECD. Permutation entropy compresses the pattern distribution of a *single* series into a scalar complexity; RECD instead builds a *multivariate hierarchy of conjunctions* and an event-driven clock.
- **Recurrence quantification analysis (RQA).** RQA quantifies recurrence structure in reconstructed phase space Marwan et al. [2007]. RECD is related in spirit—both care about returns of structure—but operates in ordinal symbol space with an explicit three-level nesting and a coupling to τ_s .
- **Transfer entropy and PID.** Transfer entropy measures directed information flow Schreiber [2000]; partial information decomposition isolates synergistic atoms Williams and Beer [2010], Lizier [2014]. Our Level 3 claim object is the irreducible residual Res_{pair} ; excess3 is a simple, falsifiable proxy, not a full PID lattice. Natural extensions replace (24) by Res_{pair} or by an axiomatic PID synergy atom when estimators permit.
- **Topological data analysis (TDA).** Persistent homology tracks multiscale topological features of point clouds Carlsson [2009]. Systemic Tau is “topological” only in the weak sense of rank geometry; it does not compute barcodes. Hybrid pipelines that feed ordinal cores into TDA functors are an open direction.

8.4 Limitations

Several limitations must be stated clearly.

1. **Proxy dependence of Level 3 (central).** The proxy `excess3` is a hybrid heuristic and is not the irreducible residual Res_{pair} . It can rise under pure pairwise (common-driver) geometries via the `Surp` term; it can agree with Res_{pair} on XOR-type higher-order joints. The coefficients $(0.6, 0.4)$ fix a hybrid class a priori (Section 4.3), not a free fit. Alternative estimators (pairwise maxent / IPF, PID atoms) may shift quantitative thresholds even when qualitative regime contrasts persist. Claims of the irreducible residual should name Res_{pair} and the window length.
2. **Calibration of λ .** The map from $|\tau_s|$ to λ can invert on some coupled systems (Section 7, Figure 3). Regime labels or multi-indicator cues may be required for stable aggregation weights; a concrete multi-cue protocol is given in Appendix A. Aggregation shares f_ℓ inherit this fragility; raw Level 3 abundance does not.
3. **Parameter choices.** Embedding (m, τ) , persistence $p_{\min}$, windows W and w , and θ_3 are operational. Sensitivity analysis is mandatory in any empirical use; the present paper reports a reference configuration used in synthetic tests.
4. **Scope of validation.** The synthetic protocol uses weakly coupled logistic maps at fixed coupling, with a binary contrast and a denser r -grid along the cascade. It supports directional predictions on proxy abundance (`excess3`) and—under ground-truth λ —on the aggregation share f_3 ; it does not exhaust the bifurcation structure of high-dimensional networks Strogatz [2001], nor prove limit theorems for $\text{Res}_{\text{pair}}(r)$ along the full Feigenbaum cascade. The `excess3` peak near r_∞ is a robust observation; the few-state vs. mixing account in Section 7.4 is a working hypothesis only.
5. **Aggregation policy, not metaphysics.** The aggregation share f_ℓ in (33) is an axiomatized design output, not a unique law of the joint symbol process and not a substitute for Res_{pair} .

9 Conclusions and Outlook

We have laid foundations for Systemic Tau and hierarchical ordinal conjunctions—with nested RECD as the operational discrete clock—as a dual relational account of critical transitions. Systemic Tau (τ_s) aggregates Kendall-type concordance across coordinates into a multivariate ordinal measure of coupling, equipped with Feigenbaum-motivated operational bands and a directed contrast $\Delta\tau_s$. The RECD constructs an intrinsic discrete clock from three levels of structural claim—coincidence (Φ_1), persistent relational structure (Φ_2), and the irreducible residual beyond pairwise models (Res_{pair})—together with a scalable proxy `excess3` that must not be identified with that residual. Regime-dependent weights $\alpha_\ell(\lambda)$ form an admissible aggregation policy that can shift the aggregation share of ΔRECD toward deeper layers near chaos; they do not create Level 3 geometry. In this language, early warning is cast as a reorganization of hierarchical ordinal structure, not merely a rise in fluctuation size—with claims, estimators, and aggregation kept distinct.

A controlled synthetic validation on coupled logistic maps supports a directional prediction for proxy abundance (`excess3`) and, under ground-truth $\lambda(r)$, for the aggregation share f_3 of ΔRECD from pre-chaotic to chaotic regimes. The same suite documents window bias for Res_{pair} , polarity inversion of $|\tau_s|$ -based cues, and a proxy peak near accumulation (observation) with only a working hypothesis for its geometry. Classical EWS remain valuable inside their theoretical domain; the present framework

is offered as a complement for high-dimensional, noisy, and nonstationary systems where relational geometry—and its depth—may carry a leading signature of criticality.

Promising directions include: (i) routine estimation of Res_{pair} and head-to-head comparison with excess3 on XOR versus common-driver synthetics; (ii) axiomatic PID synergy estimators in place of the hybrid when feasible; (iii) extensions beyond three levels or to multilayer networks; (iv) multi-cue λ as an incomplete-information estimator of the regime coordinate (Appendix A); (v) limit theorems for proxy abundance and Res_{pair} along the Feigenbaum cascade, including a proper test of the few-state vs. mixing *hypothesis* for the peak near r_∞ , and shared benchmarks against standard CSD suites. Empirical domains—from epidemiology to network physiology—provide natural testing grounds. The foundational contribution is a coherent calculus that keeps proxy and irreducible residual, and abundance and aggregation share, distinct.

A Calibration of the regime cue

This appendix records an operational protocol for constructing λ when the pure $|\tau_s|$ proxy is unreliable—as documented for the coupled logistic ensemble of Section 7—while preserving the design goal that λ increase toward chaotic or high-volatility regimes.

A.1 Design goal and polarity

Let $\mathcal{R}_{\text{chaos}}$ denote the set of regimes in which Level 3 abundance is predicted to rise (developed chaos, strong mixing, or analogous high-volatility states). The aggregation engine requires

$$\mathbb{E}[\lambda \mid \mathcal{R}_{\text{chaos}}] > \mathbb{E}[\lambda \mid \mathcal{R}_{\text{pre}}], \quad (37)$$

so that $\alpha_3(\lambda)$ in (29) elevates rather than suppresses the Level 3 term in the clock. Equation (28) supplies a one-parameter family indexed by polarity $s \in \{+1, -1\}$. The correct s is *not* universal: it encodes whether the system desynchronizes ($|\tau_s|$ falls) or remains strongly rank-coupled (or anti-coupled) as chaos develops.

Polarity diagnostic. Given labeled training windows from pre-critical and post-critical (or pre-chaos / chaos) segments, estimate $\overline{|\tau_s|}_{\text{pre}}$ and $\overline{|\tau_s|}_{\text{chaos}}$. Set

$$s = \begin{cases} -1 & \text{if } \overline{|\tau_s|}_{\text{chaos}} < \overline{|\tau_s|}_{\text{pre}}, \\ +1 & \text{otherwise.} \end{cases} \quad (38)$$

On the $d = 4$ logistic proxy of Section 7, $\overline{|\tau_s|}$ is higher pre-chaos than in chaos, so the diagnostic returns $s = -1$. Figure 3’s annotated λ_{emp} values (≈ 0.181 pre vs. ≈ 0.108 chaos) correspond to the *uncalibrated* $s = +1$ template and therefore violate (37); applying $s = -1$ restores directional consistency for that proxy. When even a polarity-corrected $|\tau_s|$ cue remains weak relative to spectral or recurrence channels, a multi-cue construction is preferred.

A.2 Multi-cue consensus

Status. The cue λ is the regime coordinate of the aggregation engine (λ, α) : the scalar that moves α_ℓ under depth monotonicity (Section 5.1). With known control parameter one sets $\lambda = \lambda(r)$. When r is latent, λ must be estimated. A single channel $\lambda_\tau(|\tau_s|)$ fails the design goal (37) whenever ordinal coupling and chaoticity are not monotone together (Section 7)—a limitation of that channel as a regime statistic, not of Systemic Tau as a relational observable.

Multi-cue λ is therefore an incomplete-information estimator of the same coordinate: a small convex consensus of independent regime signatures standard in nonlinear time series (rank coupling, spectral complexity, recurrence). It is a design object for the aggregation plane, not a dynamical law of S_t and not a substitute for Res_{pair} or proxy abundance. Naturalness criteria: channels probe distinct aspects of the transition; no channel dominates α_3 by circular construction; convex weights are fixed on a labeled holdout, not tuned to maximize f_3 on the test set.

Construction. Define a small set of normalized cues $c_j(t) \in [0, 1]$, each increasing with estimated chaoticity after its own polarity calibration:

- c_τ : monotone transform of $|\tau_s|$ with polarity s as above (e.g. a logistic map of $s(|\tau_s| - \tau_{\text{ch}})/\tau_{\text{ch}}$ onto $[0, 1]$);
- c_{spec} : spectral broadbandness (fraction of power above a mid-band cutoff, or spectral entropy);
- c_{rqa} : recurrence irregularity (e.g. $1 - \text{DET}$ or laminarity deficit from recurrence quantification Marwan et al. [2007]);
- c_{L3} : raw Level 3 *abundance* (windowed mean excess3, min–max normalized on a reference epoch), used only as a *slow* cue with small weight to avoid circular domination of α_3 by the quantity it reweights.

The consensus cue is the convex combination

$$\lambda_{\text{multi}}(t) = \left(\sum_{j \in \mathcal{J}} w_j c_j(t) \right) \cdot (1 + \gamma \log \delta), \quad w_j \geq 0, \quad \sum_{j \in \mathcal{J}} w_j = 1, \quad (39)$$

with default equal weights on $\{c_\tau, c_{\text{spec}}, c_{\text{rqa}}\}$ and optional small weight on c_{L3} ($w_{\text{L3}} \leq 0.15$) when independent spectral/RQA channels are unavailable. The Feigenbaum prefactor $(1 + \gamma \log \delta)$ is retained for continuity with (28) (role δ_{cue} only).

A.3 Fallback hierarchy

In applications we recommend the following precedence:

1. **Ground-truth control** r (or known regime labels): set $\lambda = \lambda(r)$ monotone (as in Section 7).
2. **Multi-cue consensus** (39) with polarity diagnostics on a labeled calibration segment.
3. **Single-proxy** λ_τ of (28) only after verifying (37) on held-out labeled windows.
4. **Unweighted RECD** ($\alpha_\ell \equiv 1$): report raw Φ_ℓ and excess3 contrasts without aggregation reweighting when no reliable cue exists.

This hierarchy separates two claims that the main text keeps distinct: (i) Level 3 *abundance* (excess_3 , highL_3 , or Res_{pair}) is a raw hierarchical observable and does not require λ ; (ii) Level 3 *aggregation share* f_3 under $\alpha_\ell(\lambda)$ is a regime-aware construct whose validity inherits from the quality of λ . The synthetic validation of Section 7 tests (i) unconditionally and (ii) under ground-truth $\lambda(r)$, while Figure 3 and the present appendix document why (ii) must not be claimed from an uncalibrated $|\tau_s|$ proxy on every system.

Acknowledgments

The author thanks collaborators and open-source contributors to the `systemictau` ecosystem, and acknowledges the broader complex-systems community whose work on early warning signals, ordinal patterns, and Feigenbaum universality made the present synthesis possible.

References

- José M. Amigó. *Permutation Complexity in Dynamical Systems*. Springer, Berlin, Heidelberg, 2010. doi: 10.1007/978-3-642-04084-9.
- Christoph Bandt and Bernd Pompe. Permutation entropy: a natural complexity measure for time series. *Physical Review Letters*, 88(17):174102, 2002. doi: 10.1103/PhysRevLett.88.174102.
- Carl Boettiger, Noam Ross, and Alan Hastings. Early warning signals: the charted and uncharted territories. *Theoretical Ecology*, 6:255–264, 2013. doi: 10.1007/s12080-013-0192-6.
- Gunnar Carlsson. Topology and data. *Bulletin of the American Mathematical Society*, 46(2):255–308, 2009. doi: 10.1090/S0273-0979-09-01249-X.
- Vasilis Dakos, Stephen R. Carpenter, William A. Brock, Aaron M. Ellison, Vishwesha Guttal, Anthony R. Ives, Sonia Kéfi, Valerie Livina, David A. Seekell, Egbert H. van Nes, and Marten Scheffer. Methods for detecting early warnings of critical transitions in time series illustrated using simulated ecological data. *PLOS ONE*, 7(7):e41010, 2012. doi: 10.1371/journal.pone.0041010.
- Vasilis Dakos, Stephen R. Carpenter, Egbert H. van Nes, and Marten Scheffer. Resilience indicators: prospects and limitations for early warnings of regime shifts. *Philosophical Transactions of the Royal Society B*, 370(1659):20130263, 2015. doi: 10.1098/rstb.2013.0263.
- Peter D. Ditlevsen and Sigfus J. Johnsen. Tipping points: early warning and wishful thinking. *Geophysical Research Letters*, 37(19):L19703, 2010. doi: 10.1029/2010GL044486.
- Mitchell J. Feigenbaum. Quantitative universality for a class of nonlinear transformations. *Journal of Statistical Physics*, 19(1):25–52, 1978. doi: 10.1007/BF01020332.
- Mitchell J. Feigenbaum. Universal behavior in nonlinear systems. *Physica D*, 7(1–3):16–39, 1983. doi: 10.1016/0167-2789(83)90112-4.
- Kunihiro Kaneko. Overview of coupled map lattices. *Chaos*, 2(3):279–282, 1992. doi: 10.1063/1.165869.

- Karsten Keller, Anton M. Unakafov, and Valentina A. Unakafova. Ordinal patterns, entropy, and EEG. *Entropy*, 16(12):6212–6239, 2014. doi: 10.3390/e16126212. Representative of the ordinal-pattern programme.
- Maurice G. Kendall. A new measure of rank correlation. *Biometrika*, 30(1/2):81–93, 1938. doi: 10.2307/2332226.
- Christian Kuehn. A mathematical framework for critical transitions: bifurcations, fast–slow systems and stochastic dynamics. *Physica D*, 240(12):1020–1035, 2011. doi: 10.1016/j.physd.2011.02.012.
- Joseph T. Lizier. JIDT: An information-theoretic toolkit for studying the dynamics of complex systems. *Frontiers in Robotics and AI*, 1:11, 2014. doi: 10.3389/frobt.2014.00011.
- Ronnie Mainieri and Jan Rehacek. Projective synchronization in three-dimensional chaotic systems. *Physical Review Letters*, 82(15):3042–3045, 1999. doi: 10.1103/PhysRevLett.82.3042.
- Norbert Marwan, M. Carmen Romano, Marco Thiel, and Jürgen Kurths. Recurrence plots for the analysis of complex systems. *Physics Reports*, 438(5–6):237–329, 2007. doi: 10.1016/j.physrep.2006.11.001.
- Michael McCullough, Michael Small, Thomas Stemler, and Herbert H. C. Iu. Time lagged ordinal partition networks for capturing dynamics of continuous dynamical systems. *Chaos*, 25(5):053101, 2015. doi: 10.1063/1.4919075.
- Johel Padilla-Villanueva. Unveiling systemic Tau: Redefining the fabric of time, stability, and emergent order across complex chaotic systems in the age of interdisciplinary discovery. Preprints.org, 2025a. URL <https://doi.org/10.20944/preprints202509.1428.v1>. Foundational preprint on Systemic Tau.
- Johel Padilla-Villanueva. Numerical validation of the discrete extramental clock law: Hierarchical emergence of objective time from ordinal conjunctions in chaotic systems. Preprints.org, 2025b. URL <https://doi.org/10.20944/preprints202512.2431.v1>. Nested ordinal conjunctions and synthetic RECD hierarchy on chaotic maps.
- Johel Padilla-Villanueva. systemictau: software for systemic tau and RECD analysis. Zenodo, 2026a. URL <https://doi.org/10.5281/zenodo.20576241>. Open-source implementation; archival deposit.
- Johel Padilla-Villanueva. Síntesis magna del tau sistémico. Zenodo, 2026b. URL <https://doi.org/10.5281/zenodo.20576241>. Technical synthesis of Systemic Tau thresholds, RECD, and Feigenbaum gating.
- Louis M. Pecora and Thomas L. Carroll. Synchronization in chaotic systems. *Physical Review Letters*, 64(8):821–824, 1990. doi: 10.1103/PhysRevLett.64.821.
- Marten Scheffer. *Critical Transitions in Nature and Society*. Princeton University Press, Princeton, NJ, 2012.

- Marten Scheffer, Jordi Bascompte, William A. Brock, Victor Brovkin, Stephen R. Carpenter, Vasilis Dakos, Hermann Held, Egbert H. van Nes, Max Rietkerk, and George Sugihara. Early-warning signals for critical transitions. *Nature*, 461:53–59, 2009. doi: 10.1038/nature08227.
- Thomas Schreiber. Measuring information transfer. *Physical Review Letters*, 85(2):461–464, 2000. doi: 10.1103/PhysRevLett.85.461.
- Steven H. Strogatz. Exploring complex networks. *Nature*, 410:268–276, 2001. doi: 10.1038/35065725.
- James Theiler, Stephen Eubank, André Longtin, Bryan Galdrikian, and J. Doyne Farmer. Testing for nonlinearity in time series: the method of surrogate data. *Physica D*, 58(1–4):77–94, 1992. doi: 10.1016/0167-2789(92)90102-S.
- Paul L. Williams and Randall D. Beer. Nonnegative decomposition of multivariate information. *arXiv preprint arXiv:1004.2515*, 2010.
- Alan Wolf, Jack B. Swift, Harry L. Swinney, and John A. Vastano. Determining Lyapunov exponents from a time series. *Physica D*, 16(3):285–317, 1985. doi: 10.1016/0167-2789(85)90011-9.